Chemistry 159 Exam I October 17, 2017

Student Name: _____

Recitation Section Number: ____

Recitation Instructor: _____

The exam has 25 questions for credit and an additional one to check the color of your exam booklet. Please answer all 26 questions **on the OpScan sheet.** There is no penalty for guessing. If you finish early, please do not disturb your fellow students. A proctor will check your picture-ID, OpScan form and signature *during* the exam. Exam scores and an exam curve will be posted as soon as possible.

ON THE OpScan FORM: (Use a Number 2 Pencil or Darker)

(1) SIGN your name across the TOP of the form.

(2) code only the following information: [**blacken** circles]

(a) Your Name . . . LAST NAME FIRST
[**blank** *space between last and first names*]

(b) Your RUTGERS ID NUMBER
[Start under Box **A** and continue to Box **I**]

(c) Your RECITATION SECTION NUMBER (Boxes **K & L**)

(d) Your EXAM FORM NUMBER (Box **P**)

See last pages of exam for physical constants, periodic table, and other information.

1	What is the frequency of light emitted when an electron jumps from state n = 3 to the ground state in a hydrogen atom? A. $3.29 \times 10^{15}\ s^{-1}$ B. $3.65 \times 10^{14}\ s^{-1}$ C. $2.92 \times 10^{15}\ s^{-1}$ D. $1.51 \times 10^{15}\ s^{-1}$ E. $5.72 \times 10^{15}\ s^{-1}$
2	Which of the following atoms or ions has exactly three singly occupied orbitals? X. Sc Y Mn^{2+} Z. Co A. X only B. Y only C. Z only D. X and Z only E. Y and Z only
3	How much energy is required to excite an electron from energy level n = 2 to energy level n = 5 in the hydrogen atom? A. 4.58×10^{-19} J B. 1.56×10^{-18} J C. 1.94×10^{-18} J D. 3.03×10^{-19} J E. 1.82×10^{-19} J
4	Select the atom/ion that has the incorrect configuration listed next to it. A. P: $[Ne]3s^2 3p^3$ B. Ag: $[Kr]5s^1 4d^{10}$ C. Sr^{2+}: [Kr] D. Ti^{2+}: $[Ar]4s^2$ E. Y: [Kr] $5s^2 4d^1$
5	Which of the following are true statements? X. The mass of 1 mol of carbon-12 is exactly 12 g. Y. Carbon contains at least one natural isotope heavier than carbon-12. Z. The mass of one atom of carbon-12 is 1.99×10^{-23} g A. X only B. Y only C. Z only D. X and Z only E. X, Y, and Z

6 $^{41}_{19}K$ has mass 40.9615 amu. What amount of neutrons is in a sample of $^{41}_{19}K$ having a mass of 10.0 g?

A. 10.9 moles
B. 5.37 moles
C. 31.0 moles
D. 24.2 moles
E. 18.6 moles

7 A homogeneous mixture was prepared by mixing 0.15g A, 112.3g B and the 24.55g C. The volume is found to be 85.80 cm^3. What is the density of the mixture to correct number of significant digits?

A. 1.5967 $\frac{g}{cm^3}$
B. 1.59674 $\frac{g}{cm^3}$
C. 1.6 $\frac{g}{cm^3}$
D. 1.60 $\frac{g}{cm^3}$
E. 1.597 $\frac{g}{cm^3}$

8 Which one of the following sets of quantum numbers represents an electron in a 4d orbital?

	n	l	ml	ms
A.	4	1	−1	+½
B.	4	2	−3	−½
C.	4	3	−2	+½
D.	4	0	0	+½
E.	4	2	0	−½

9 Determine the volume of hexane that contains 5.33×10^{22} molecules of hexane. The density of hexane is 0.6548 g/mL and its molar mass is 86.17 g/mol.
A. 8.59 mL
B. 13.5 mL
C. 7.40 mL
D. 12.4 mL
E. 11.6 mL

10 Determine the name for TiCO3.

A. titanium (II) carbonate
B. titanium carbide
C. titanium carbonite
D. titanium (II) carbonite
E. titanium (I) carbonate

11 The threshold frequency to eject an electron from the surface of a metal is $1.95 \times 10^{14}\ s^{-1}$. What is the kinetic energy of an electron ejected from the surface of the same metal if it is bombarded with a light with a wavelength of 358 nm?

A. 5.55×10^{-19} J
B. 4.26×10^{-19} J
C. 1.29×10^{-19} J
D. 6.84×10^{-19} J
E. 8.13×10^{-19} J

12 Three individuals each made 4 measurements of the length of a steel rod. The actual length of the rod is 7.54 m. The results of the measurements are shown below:

Individual	Length (m)
X	7.24, 7.89, 7.60, 7.22
Y	7.31, 7.32, 7.30, 7.32
Z	7.52, 7.56, 7.58, 7.49

Which of the following statements represents the results shown above?

A. Y was most accurate and Z is the most precise
B. X and Z were both equally accurate and precise
C. X was most accurate and Y was most precise
D. Z was the most accurate and X was most precise
E. Z was the most accurate and Y is the most precise

13 Combustion analysis of 63.8 mg of a C, H, and O containing compound produced 145.0 mg of CO_2 and 59.38 mg of H_2O. What is the empirical formula for the compound?

A. C_5H_2O
B. CHO
C. C_3H_6O
D. C_3H_7O
E. C_6HO_3

14 What is the de Broglie wavelength for a neutron traveling at 700 m/s.?
Mass of neutron: 1.675×10^{-27} kg

A. 266 nm
B. 37.4 nm
C. 0.377 nm
D. 1.77 nm
E. 0.565 nm

15	In the n = 4 principal shell, what is the maximum number of electrons that could have ms = +½ ? A. 1 B. 4 C. 8 D. 16 E. 32
16	A certain element has just one natural isotope. If 6.4 ng of this element contains 3.04×10^{13} atoms identify the element. A. beryllium B. sodium C. gallium D. strontium E. iodine
17	What is the dollar value of a cube of gold measuring 2.00 inches on a side? Price of gold: \$1366 per troy ounce 31.1 g = 1 troy ounce Density of gold: 19.3 g/cm^3 2.54 cm = 1 inch A. \$$1.11 \times 10^5$ B. \$$2.41 \times 10^7$ C. \$$2.98 \times 10^4$ D. \$$1.07 \times 10^8$ E. \$$5.96 \times 10^6$
18	What would be the volume of a block of iron that contains 7.93×10^{24} atoms of iron? Density of iron = 7.874 $\frac{g}{cm^3}$ A. 735 cm^3 B. 281 cm^3 C. 1.67 cm^3 D. 54.7 cm^3 E. 93.4 cm^3
19	How many moles of photons with a wavelength of 679 nm would need to strike a material in order for it to absorb 54.7 kJ of energy? A. 0.192 mole B. 0.310 mole C. 0.259 mole D. 0.671 mole E. 0.127 mole

20 Use the data given below to construct a Born-Haber cycle to determine the lattice energy of CaO.

	$\Delta H°$(kJ)
$Ca(s) \rightarrow Ca(g)$	193
$Ca(g) \rightarrow Ca^+(g) + e^-$	590
$Ca^+(g) \rightarrow Ca^{2+}(g) + e^-$	1010
$2\ O(g) \rightarrow O_2(g)$	-498
$O(g) + e^- \rightarrow O^-(g)$	-141
$O^-(g) + e^- \rightarrow O^{2-}(g)$	878
$Ca(s) + \frac{1}{2}O_2(g) \rightarrow CaO(s)$	-635

A. -3414 kJ
B. +1397 kJ
C. -2667 kJ
D. +3028 kJ
E. -2144 kJ

21 In each of the following pairs, select the species with the smaller radius.

W:	K^+	Cl^-
X:	Br	Br^-
Y:	Ca	Mg^{2+}
Z:	S	F

	W	X	Y	Z
A.	K^+	Br	Mg^{2+}	F
B.	Cl^-	Br^-	Mg^{2+}	F
C.	K^+	Br	Ca	S
D.	K^+	Br^-	Mg^{2+}	F
E.	Cl^-	Br	Ca	S

22 Determine the molecular formula of a compound that is 49.48% carbon, 5.19% hydrogen, 28.85% nitrogen, and 16.48% oxygen. The molecular weight is 194.19 g/mol.

A. $C_8H_{12}N_4O_2$
B. $C_4H_5N_2O$
C. $C_8H_{10}N_4O_2$
D. $C_8H_{10}N_2O$
E. $C_8H_{12}N_4O_2$

23 Which of the following are intensive properties?

X. density Y. volume Z. boiling point

A. X only
B. Y only
C. Z only
D. X and Z only
E. X, Y, and Z only

24	Rank the following according to increasing lattice energy. (Lowest first, highest last) MgO, $CaCl_2$, KBr, KCl, $MgCl_2$ A. $MgO < CaCl_2 < MgCl_2 < KBr < KCl$ B. $KBr < KCl < CaCl_2 < MgCl_2 < MgO$ C. $KBr < CaCl_2 < KCl < MgO < MgCl_2$ D. $KCl < MgCl_2 < KBr < CaCl_2 < MgO$ E. $MgO < MgCl_2 < KBr < KCl < CaCl_2$
25	Lithium forms compounds which are used in dry cells and storage batteries and in high-temperature lubricants. It has two naturally occurring isotopes, ^6Li (isotopic mass = 6.015121 amu) and ^7Li (isotopic mass = 7.016003 amu). Lithium has an atomic mass of 6.9409 amu. How many atoms of ^6Li are in a naturally occurring sample of lithium that contains a total of 4.00×10^{20} atoms? A. 3.00×10^{19} B. 4.60×10^{19} C. 3.80×10^{19} D. 5.00×10^{19} E. 2.20×10^{19}
26.	What is the color of your exam? A. White B. Yellow C. Blue D. Green

Constants and Formulas

1 N = 1 kg m/s^2

1 J = 1 kg m^2/s^2

1 L·atm = 101.325 J

T(°F) = 1.8 T(°C) + 32

T(K) = T(°C) + 273.15

R = 0.08206 L atm mol^{-1} K^{-1} = 8.3145 J mol^{-1} K^{-1}

N_A = 6.022 x 10^{23} mol^{-1}

h = 6.626 x 10^{-34} J·s

c = 2.998 x 10^8 m/s

g = 9.80665 m/s^2

$E \propto \frac{Q_1 Q_2}{r}$ $F \propto \frac{Q_1 Q_2}{r^2}$

$E = h\nu$ $c = \lambda\nu$ $\lambda = \frac{h}{mv}$

$KE = \frac{1}{2}mv^2$ $KE = h\nu - h\nu_0 = h\nu - W$ where W = binding energy $\Delta x \cdot m\Delta v \geq \frac{h}{4\pi}$

$E_n = -B\left(\frac{1}{n^2}\right)$ $\Delta E = -B\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$ $|\Delta E| = h\nu$ $B = 2.179 \times 10^{-18}$ J

$\Delta E = q + w$ $w = -P\Delta V$ $H = E + PV$

$q = m \cdot C \cdot \Delta T$

$$\Delta H°_{rxn} = \sum n_P \Delta H°_f(products) - \sum n_R \Delta H°_f(reactants)$$

$$\Delta H° = \sum BE(broken) - \sum BE(formed)$$

$u_{rms} = \sqrt{\frac{3RT}{\mathcal{M}}}$ $\frac{r_1}{r_2} = \frac{t_2}{t_1} = \sqrt{\frac{\mathcal{M}_2}{\mathcal{M}_1}}$ $KE = \frac{1}{2}mu^2 = \frac{3}{2}RT$

PV = nRT $P = dgh$ $P_i = \chi_i P_{total}$

$$\%\ yield = \frac{Actual\ yield}{Theoretical\ yield} \times 100$$

Soluble	Exceptions	Insoluble
Li^+, Na^+, K^+, Rb^+, Cs^+, NH_4^+	none	
NO_3^-, CH_3COO^-, ClO_3^-	none	
Cl^-, Br^-, I^-	Ag^+, Hg_2^{2+}, Pb^{2+}	
SO_4^{2-}	Ca^{2+}, Sr^{2+}, Ba^{2+}, Ag^+, Hg_2^{2+}, Pb^{2+}	
	Li^+, Na^+, K^+, Rb^+, Cs^+, Ba^{2+}	OH^-
	Li^+, Na^+, K^+, Rb^+, Cs^+, NH_4^+	CO_3^{2-}, PO_4^{3-}, S^{2-}, CrO_4^{2-}

Periodic Table of the Elements

1 I A																	18 VIII A
1 **H** Hydrogen 1.008	2 II A											13 III A	14 IV A	15 V A	16 VI A	17 VII A	2 **He** Helium 4.00
3 **Li** Lithium 6.94	4 **Be** Beryllium 9.01											5 **B** Boron 10.81	6 **C** Carbon 12.01	7 **N** Nitrogen 14.01	8 **O** Oxygen 16.00	9 **F** Fluorine 19.00	10 **Ne** Neon 20.18
11 **Na** Sodium 22.99	12 **Mg** Magnesium 24.30	3 III B	4 IV B	5 V B	6 VI B	7 VII B	8	9 VIII B	10	11 I B	12 II B	13 **Al** Aluminum 26.98	14 **Si** Silicon 28.09	15 **P** Phosphorus 30.97	16 **S** Sulfur 32.06	17 **Cl** Chlorine 35.45	18 **Ar** Argon 39.95
19 **K** Potassium 39.10	20 **Ca** Calcium 40.08	21 **Sc** Scandium 44.96	22 **Ti** Titanium 47.90	23 **V** Vanadium 50.94	24 **Cr** Chromium 52.00	25 **Mn** Manganese 54.94	26 **Fe** Iron 55.85	27 **Co** Cobalt 58.93	28 **Ni** Nickel 58.70	29 **Cu** Copper 63.55	30 **Zn** Zinc 65.38	31 **Ga** Gallium 69.72	32 **Ge** Germanium 72.59	33 **As** Arsenic 74.92	34 **Se** Selenium 78.96	35 **Br** Bromine 79.90	36 **Kr** Krypton 83.80
37 **Rb** Rubidium 85.47	38 **Sr** Strontium 87.62	39 **Y** Yttrium 88.91	40 **Zr** Zirconium 91.22	41 **Nb** Niobium 92.91	42 **Mo** Molybdenum 95.94	43 **Tc** Technetium 98.91	44 **Ru** Ruthenium 101.07	45 **Rh** Rhodium 102.91	46 **Pd** Palladium 106.4	47 **Ag** Silver 107.87	48 **Cd** Cadmium 112.41	49 **In** Indium 114.82	50 **Sn** Tin 118.69	51 **Sb** Antimony 121.75	52 **Te** Tellurium 127.60	53 **I** Iodine 126.90	54 **Xe** Xenon 131.30
55 **Cs** Cesium 132.91	56 **Ba** Barium 137.33	57* **La** Lanthanum 138.91	72 **Hf** Hafnium 178.49	73 **Ta** Tantalum 180.95	74 **W** Tungsten 183.85	75 **Re** Rhenium 186.21	76 **Os** Osmium 190.2	77 **Ir** Iridium 192.22	78 **Pt** Platinum 195.09	79 **Au** Gold 196.97	80 **Hg** Mercury 200.59	81 **Tl** Thallium 204.37	82 **Pb** Lead 207.2	83 **Bi** Bismuth 208.98	84 **Po** Polonium (209)	85 **At** Astatine (210)	86 **Rn** Radon (222)
87 **Fr** Francium (223)	88 **Ra** Radium 226.03	89† **Ac** Actinium 227.03	104 **Unq** Unnilquadium (261)	105 **Unp** Unnilpentium (262)	106 **Unh** Unnilhexium (263)	107 **Uns** Unnilseptium (262)	108 **Uno** Unniloctium (265)	109 **Une** Unnilennium (267)									

Atomic number → 6
C ← Symbol
Carbon ← Name
Atomic mass → 12.01

Lanthanide Series*	58 **Ce** Cerium 140.12	59 **Pr** Praseodymium 140.91	60 **Nd** Neodymium 144.24	61 **Pm** Promethium (145)	62 **Sm** Samarium 150.4	63 **Eu** Europium 151.96	64 **Gd** Gadolinium 157.25	65 **Tb** Terbium 158.93	66 **Dy** Dysprosium 162.50	67 **Ho** Holmium 164.93	68 **Er** Erbium 167.26	69 **Tm** Thulium 168.93	70 **Yb** Ytterbium 173.04	71 **Lu** Lutetium 174.97
Actinide Series†	90 **Th** Thorium 232.04	91 **Pa** Protactinium 231.04	92 **U** Uranium 238.03	93 **Np** Neptunium 237.05	94 **Pu** Plutonium (244)	95 **Am** Americium (243)	96 **Cm** Curium (247)	97 **Bk** Berkelium (247)	98 **Cf** Californium (251)	99 **Es** Einsteinium (252)	100 **Fm** Fermium (257)	101 **Md** Mendelevium (258)	102 **No** Nobelium (259)	103 **Lr** Lawrencium (262)

159 Fall 17, Exam I

KEY V4

NAME (Last, First, M.I.)

SEX: (M) (F)

GRADE OR EDUC: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

200 ITEM ANSWER SHEET

SIDE 1

INSTRUCTIONS

PLEASE USE A NO. 2 PENCIL ONLY
MAKE HEAVY BLACK MARKS THAT FILL THE CIRCLE COMPLETELY
DO NOT MAKE ANY STRAY MARKS ON THIS ANSWER SHEET
MAKE ALL ERASURES CLEANLY

EXAMPLES: PROPER MARK ● IMPROPER MARKS ◒ ⊘ ⊗

1 B	11 C	21 A	31 B	41 B
2 D	12 E	22 C	32 B	42 B
3 A	13 C	23 D	33 B	43 B
4 D	14 D	24 B	34 B	44 B
5 E	15 C	25 A	35 B	45 B
6 A	16 E	26 D	36 B	46 B
7 E	17 A	27 B	37 B	47 B
8 E	18 E	28 B	38 B	48 B
9 E	19 B	29 B	39 B	49 B
10 A	20 A	30 B	40 B	50 B
51 B	61 B	71 B	81 B	91 B
52 B	62 B	72 B	82 B	92 B
53 B	63 B	73 B	83 B	93 B
54 B	64 B	74 B	84 B	94 B
55 B	65 B	75 B	85 B	95 B
56 B	66 B	76 B	86 B	96 B
57 B	67 B	77 B	87 B	97 B
58 B	68 B	78 B	88 B	98 B
59 B	69 B	79 B	89 B	99 B
60 B	70 B	80 B	90 B	100 B

BIRTH DATE

MO.	DAY	YR.
Jan.		
Feb.		
Mar.	0 0	0 0
Apr.	1 1	1 1
May	2 2	2 2
Jun.	3 3	3 3
Jul.	4 4	4 4
Aug.	5 5	5 5
Sep.	6 6	6 6
Oct.	7 7	7 7
Nov.	8 8	8 8
Dec.	9 9	9 9

IDENTIFICATION NUMBER

A	B	C	D	E	F	G	H	I	J
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

SPECIAL CODES

K	L	M	N	O	P
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

P: 4

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Chemistry 159 Exam II November 21, 2017

Student Name Student Name: _____

Recitation Section Number: ____

Recitation Instructor: _____

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(c) Your RECITATION SECTION NUMBER (Boxes **K & L**)

(d) Your EXAM FORM NUMBER (Box **P**)

See last pages of exam for physical constants, periodic table, and other information.

1 Balance the following equation using the smallest set of whole numbers, then add together the coefficients. Don't forget to count coefficients of one. The sum of the coefficients is:

__ SF_4 + __ H_2O → __ H_2SO_3 + __ HF

A. 4
B. 6
C. 7
D. 9
E. None of the answers is correct.

2 Balance the chemical equation given below, and calculate the volume of nitrogen monoxide gas produced when 8.00 g of ammonia is reacted with 8.00 g of oxygen at 25°C? The density of nitrogen monoxide at 25°C is 1.23 g/L.

_____ $NH_3(g)$ + _____ $O_2(g)$ →_____ $NO(g)$ + _____ $H_2O(l)$

A) 4.88 L
B) 7.38 L
C) 11.5 L
D) 17.3 L
E) 19.2 L

3 Which bond is most polar?

A. C—C
B. H—C
C. N—F
D. Si—F
E. P—F

4 A student dissolved 5.00 g of $Co(NO_3)_2$ in enough water to make 100. mL of stock solution. He took 4.00 mL of the stock solution and then diluted it with water to give 275. mL of a final solution. How many grams of NO3- ion are there in the final solution?

A) 0.0 247 g
B) 0.0 493 g
C) 0.0 678 g
D) 0. 136 g
E. 0.475 g

5 Which compound or ion is correctly associated with its shape?

A.	NH_3	Trigonal planar
B.	SO_2	Linear
C.	SF_4	Tetrahedral
D.	ClO_4^-	Seesaw
E.	BrF_5	Square pyramidal

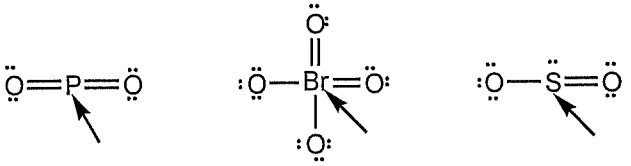
6 In the reaction: $BF_3 + F^- \rightarrow BF_4^-$, what happens to the hybridization on boron and the F-B-F bond angle?

A. Hybridization is unchanged and the bond angle decreases.
B. Hybridization is unchanged and the bond angle is unchanged.
C. Hybridization changes and the bond angle remains the same.
D. Hybridization changes and the bond angle increases.
E. Hybridization changes and the bond angle decreases.

7 Based on the solubility rules, which of these processes will occur if solutions of $CuSO_4(aq)$ and $BaCl_2(aq)$ are mixed?

A. $CuCl_2$ will precipitate; Ba^{2+} and SO_4^{2-} are spectator ions.
B. $CuSO_4$ will precipitate; Ba^{2+} and Cl^- are spectator ions.
C. $BaSO_4$ will precipitate; Cu^{2+} and Cl^- are spectator ions.
D. $BaCl_2$ will precipitate; Cu^{2+} and SO_4^{2-} are spectator ions.
E. No precipitate will form.

8 What are the formal charges on the indicated central atoms in each of the following molecules or ions?



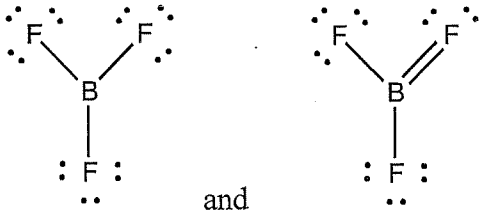
	P	Br	S
A.	+1	+1	+1
B.	0	+1	−1
C.	+1	0	+1
D.	-1	+1	0
E.	−1	−1	−1

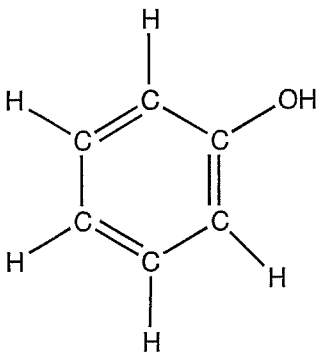
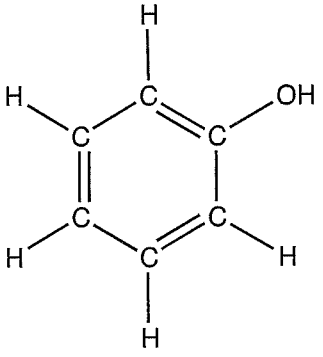
9 Which of the following molecules are polar?

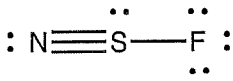
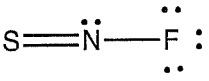
W. PCl_5 X. PCl_3 Y. XeF_2 Z. SF_2

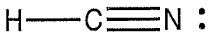
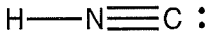
A. W and X
B. Y and Z
C. W and Y
D. X and Z
E. X and Y

10 Which of the following pairs represent resonance structures?

I.  and

II.  and 

III.  and 

IV.  and 

A. I, II, III and IV
B. II and IV
C. I and II
D. II and III
E. III and IV

11 Use the molecular orbital theory shown to determine which of the following is paramagnetic.

A. O_2^{2-}
B. Ne_2^{2+}
C. O_2^{2+}
D. F_2^{2+}
E. None of the above are paramagnetic

12 Given the equation: $Br_2 + 5F_2 \rightarrow 2BrF_5$

Suppose we start with 0.20 moles of Br_2 and 0.80 moles of F2, and 0.25 moles of BrF_5 is formed. What is the percent yield of this reaction?

A. 62%
B. 25%
C. 50%
D. 78%
E. 33%

13 Element X reacts with oxygen: $4X + 3O_2 \rightarrow 2X_2O_3$

If 30.0 g of X reacts completely with 0.300 mol of O_2, what is the molar mass of X?

A. 52.0 $\frac{g}{mol}$

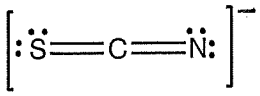
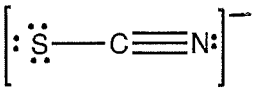
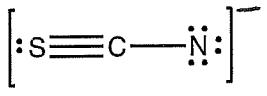
B. 75.0 $\frac{g}{mol}$

C. 31.0 $\frac{g}{mol}$

D. 88.9 $\frac{g}{mol}$

E. 122 $\frac{g}{mol}$

14 Which is the correct order of "best to worst" resonance structures for SCN^-.
Note: nitrogen is more electronegative than sulfur

X.  Y.  Z. 

	Best	Middle	Worst
A.	X	Y	Z
B.	X	Z	Y
C.	Y	X	Z
D.	Y	Z	X
E.	Z	Y	X

15	What is the molecular geometry of $XeOF_2$? A. trigonal pyramidal B. tetrahedral C. T-shaped D. trigonal planar E. bent
16	Rank the following compounds or ions according to increasing lone pairs around the central atom: W. KrF_2 X. SF_4 Y. ClO_4^- Z. BrF_3 A. X < Z < W < Y B. Y < X < Z < W C. Y < Z < W < X D. W < Y < Z < X E. Z < X < Y < W
17	A 0.500 mol sample of C_3H_7X has mass of 31.0 g. What is the identity of element X? A. fluorine B. chlorine C. bromine D. oxygen E. iodine
18	Give the **complete ionic equation** for the reaction (if any) that occurs when aqueous solutions of lithium sulfide and copper (II) nitrate are mixed. A) $Li^+(aq) + SO_4^{2-}(aq) + Cu^+(aq) + NO_3^-(aq) \rightarrow CuS(s) + Li^+(aq) + NO_3^-(aq)$ B) $Li^+(aq) + S^-(aq) + Cu^+(aq) + NO_3^-(aq) \rightarrow CuS(s) + LiNO_3(aq)$ C) $2\ Li^+(aq) + S^{2-}(aq) + Cu^{2+}(aq) + 2\ NO_3^-(aq) \rightarrow Cu^{2+}(aq) + S^{2-}(aq) + 2\ LiNO_3(s)$ D) $2\ Li^+(aq) + S^{2-}(aq) + Cu^{2+}(aq) + 2\ NO_3^-(aq) \rightarrow CuS(s) + 2\ Li^+(aq) + 2\ NO_3^-(aq)$ E) No reaction occurs.
19	How many sigma and pi bonds are present in a Lewis structure of SO_4^{2-} that minimizes formal charge? A. 4 sigma, 2 pi B. 4 sigma, 0 pi C. 2 sigma, 2 pi D. 3 sigma, 3 pi E. 5 sigma, 1 pi

20	Consider the reaction: $4NO + 5O_2 \rightarrow 4NO_2 + 6H_2O$ How many molecules of O_2 are required to react with 15.0 g of NO? A. 3.01×10^{23} molecules O_2 B. 2.51×10^{23} molecules O_2 C. 3.76×10^{23} molecules O_2 D. 1.74×10^{23} molecules O_2 E. 6.67×10^{23} molecules O_2
21	Magnesium (used in the manufacture of light alloys) reacts with iron (III) chloride to form magnesium chloride and iron. A mixture of 41.0 g of magnesium and 175 g of iron (III) chloride is allowed to react. Identify the limiting reactant and determine the mass of the excess reactant present in the vessel when the reaction is complete. $3Mg(s) + 2FeCl_3(aq) \rightarrow 3MgCl_2(aq) + 2Fe(s)$ A. Limiting reactant is Mg; 67 g of $FeCl_3$ remain. B. Limiting reactant is Mg; 134 g of $FeCl_3$ remain. C. Limiting reactant is Mg; 104 g of $FeCl_3$ remain D. Limiting reactant is $FeCl_3$; 2 g of Mg remain E. Limiting reactant is $FeCl_3$; 87 g of Mg remain.
22	When ethyne, C_2H_2, reacts with O_2 in a combustion reaction, the products of the combustion are CO_2 and H_2O Which combination of reactants would react completely with nothing left over of either reactant? A. 0.500 mol C_2H_2 + 0.750 mol O_2 B. 0.600 mol C_2H_2 + 1.20 mol O_2 C. 0.200 mol C_2H_2 + 0.600 mol O_2 D. 0.600 mol C_2H_2 + 2.00 mol O_2 E. 1.40 mol C_2H_2 + 3.50 mol O_2
23	The titration of 25.0 mL of an unknown concentration H_2SO_4 solution requires 83.6 mL of 0.12 M LiOH solution. What is the concentration of the H_2SO_4 solution (in M)? A) 0.20 M B) 0.40 M C) 0.10 M D) 0.36 M E) 0.25 M

24	Analyzing the three equivalent resonance structures for NO_3^-, the average formal charge on each oxygen atom is: A. -1 B. -2 C. -1/3 D. -2/3 E. -3
25	Which orbitals overlap in forming the C—N sigma bond in CH_2NH? A. C(sp3) – N(sp3) B. C(sp2) – N(sp3) C. C(sp3) – N(sp2) D. C(sp2) – N(sp2) E. C(2p) – N(2p)
26.	What is the color of your exam? A. White B. Yellow C. Blue D. Green

WebElements: the periodic table on the world-wide web

www.webelements.com

1	2		3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
hydrogen 1 **H** 1.0079																		helium 2 **He** 4.0026
lithium 3 **Li** 6.941	beryllium 4 **Be** 9.0122												boron 5 **B** 10.811	carbon 6 **C** 12.011	nitrogen 7 **N** 14.007	oxygen 8 **O** 15.999	fluorine 9 **F** 18.998	neon 10 **Ne** 20.180
sodium 11 **Na** 22.990	magnesium 12 **Mg** 24.305												aluminium 13 **Al** 26.982	silicon 14 **Si** 28.086	phosphorus 15 **P** 30.974	sulfur 16 **S** 32.065	chlorine 17 **Cl** 35.453	argon 18 **Ar** 39.948
potassium 19 **K** 39.098	calcium 20 **Ca** 40.078		scandium 21 **Sc** 44.956	titanium 22 **Ti** 47.867	vanadium 23 **V** 50.942	chromium 24 **Cr** 51.996	manganese 25 **Mn** 54.938	iron 26 **Fe** 55.845	cobalt 27 **Co** 58.933	nickel 28 **Ni** 58.693	copper 29 **Cu** 63.546	zinc 30 **Zn** 65.38	gallium 31 **Ga** 69.723	germanium 32 **Ge** 72.61	arsenic 33 **As** 74.922	selenium 34 **Se** 78.96	bromine 35 **Br** 79.904	krypton 36 **Kr** 83.80
rubidium 37 **Rb** 85.468	strontium 38 **Sr** 87.62		yttrium 39 **Y** 88.906	zirconium 40 **Zr** 91.224	niobium 41 **Nb** 92.906	molybdenum 42 **Mo** 95.96	technetium 43 **Tc** [98]	ruthenium 44 **Ru** 101.07	rhodium 45 **Rh** 102.91	palladium 46 **Pd** 106.42	silver 47 **Ag** 107.87	cadmium 48 **Cd** 112.41	indium 49 **In** 114.82	tin 50 **Sn** 118.71	antimony 51 **Sb** 121.76	tellurium 52 **Te** 127.60	iodine 53 **I** 126.90	xenon 54 **Xe** 131.29
caesium 55 **Cs** 132.91	barium 56 **Ba** 137.33	57-70 *	lutetium 71 **Lu** 174.97	hafnium 72 **Hf** 178.49	tantalum 73 **Ta** 180.95	tungsten 74 **W** 183.84	rhenium 75 **Re** 186.21	osmium 76 **Os** 190.23	iridium 77 **Ir** 192.22	platinum 78 **Pt** 195.08	gold 79 **Au** 196.97	mercury 80 **Hg** 200.59	thallium 81 **Tl** 204.38	lead 82 **Pb** 207.2	bismuth 83 **Bi** 208.98	polonium 84 **Po** [209]	astatine 85 **At** [210]	radon 86 **Rn** [222]
francium 87 **Fr** [223]	radium 88 **Ra** [226]	89-102 **	lawrencium 103 **Lr** [262]	rutherfordium 104 **Rf** [267]	dubnium 105 **Db** [268]	seaborgium 106 **Sg** [271]	bohrium 107 **Bh** [272]	hassium 108 **Hs** [270]	meitnerium 109 **Mt** [276]	darmstadtium 110 **Ds** [281]	roentgenium 111 **Rg** [280]	ununbium 112 **Uub** [285]	ununtrium 113 **Uut** [284]	ununquadium 114 **Uuq** [289]	ununpentium 115 **Uup** [288]	ununhexium 116 **Uuh** [293]	ununseptium 117 —	ununoctium 118 **Uuo** [294]

Key:

element name
atomic number
symbol
atomic weight (mean relative mass)

*lanthanoids	lanthanum 57 **La** 138.91	cerium 58 **Ce** 140.12	praseodymium 59 **Pr** 140.91	neodymium 60 **Nd** 144.24	promethium 61 **Pm** [145]	samarium 62 **Sm** 150.36	europium 63 **Eu** 151.96	gadolinium 64 **Gd** 157.25	terbium 65 **Tb** 158.93	dysprosium 66 **Dy** 162.50	holmium 67 **Ho** 164.93	erbium 68 **Er** 167.26	thulium 69 **Tm** 168.93	ytterbium 70 **Yb** 173.06
**actinoids	actinium 89 **Ac** [227]	thorium 90 **Th** 232.04	protactinium 91 **Pa** 231.04	uranium 92 **U** 238.03	neptunium 93 **Np** [237]	plutonium 94 **Pu** [244]	americium 95 **Am** [243]	curium 96 **Cm** [247]	berkelium 97 **Bk** [247]	californium 98 **Cf** [251]	einsteinium 99 **Es** [252]	fermium 100 **Fm** [257]	mendelevium 101 **Md** [258]	nobelium 102 **No** [259]

Symbols and names: the symbols and names of the elements, and their spellings are those recommended by the International Union of Pure and Applied Chemistry (IUPAC - http://www.iupac.org/). Names have yet to be proposed for the most recently discovered elements beyond 112 and so those used here are IUPAC's temporary systematic names. In the USA and some other countries, the spellings **aluminum** and **cesium** are normal while in the UK and elsewhere the common spelling is **sulphur**.
Group labels: the numeric system (1–18) used here is the current IUPAC convention.
Atomic weights (mean relative masses): Apart from the heaviest elements, these are the IUPAC 2007 values and given to 5 significant figures. Elements for which the atomic weight is given within square brackets have no stable nuclides and are represented by the element's longest lived isotope reported at the time of writing.
©2007 Dr Mark J Winter [WebElements Ltd and University of Sheffield. webelements@sheffield.ac.uk]. All rights reserved. For updates to this table see http://www.webelements.com/nexus/PrintablePeriodicTable (Version date: 21 September 2007).

Name _____ Section Number _____

POSSIBLY USEFUL INFORMATION

$N_A = 6.022 \times 10^{23}\ mol^{-1}$ | $T(°F) = (1.8T(°C)) + 32$ | $T(K) = T(°C) + 273.15$

$2.54\ cm = 1\ in$ | $V = (4/3)\pi r^3$ | $V = \pi r^2 h$

$1\ L = 1000\ cm^3$ | $V = \ell wh$ | $V = Ah$

$A = \pi r^2$ | $A = \ell w$ | $A = 4\pi r^2$

$C = 2\pi r$ | $PV = nRT$ | $P = F/A$

$F = ma$ | $P = gdh$ | $1\ Pa = 1\ N/m^2$

$1\ N = 1\ kg\ m/s^2$ | $g = 9.80665\ m/s^2$ | STP is 0°C, 1 atm.

$1\ J = 1\ kg\ m^2/s^2$ | $1\ L\ atm = 101.325\ J$ | $n = N / N_A$

$R = 8.3145\ J/mol\ K$ | $R = 0.08206\ L\ atm/mol\ K$ | $n = m / M$

[760 mmHg = 1 atm = 760 Torr = 101.325 kPa = 1.01325 bar] | $6.022 \times 10^{23}\ amu = 1\ g$

$u_{rms} = \sqrt{\frac{3RT}{M}}$ | $(E_k)_{average} = (3/2)(R / N_A)T$ | rate $\propto u_{rms}$ | $\frac{r_1}{r_2} = \frac{t_2}{t_1} = \sqrt{\frac{M_2}{M_1}}$

$E = (3/2)nRT$ | $q = C\Delta T$ | $q = mc\Delta T$

$\Delta E = q + w$ | $w = -P_{external}\Delta V$ | $H = E + PV$

$\Delta H = \Delta E + \Delta(PV)$ | $\Delta H = \Delta E + P\Delta V$ | $\Delta H = \Delta E + V\Delta P$

$\Delta X = X_{final} - X_{initial}$ | $h\nu = \phi + (1/2)mu^2$ | $\lambda = h / mc$

$R = 1.097 \times 10^7\ m^{-1}$ | $h = 6.626 \times 10^{-34}\ J\ s$ | $c = 3.00 \times 10^8\ m/s$

$E = h\nu$ | $E_n = -R\,h\,c\,Z^2 / n^2$ | $\lambda = h / mu$

$(\Delta x)(m\Delta u) \geq (h/4\pi)$ | $R\,h\,c = 2.18 \times 10^{-18}\ J$ | $c = \lambda\nu$

1 gallon = 4 qt | 1.057 qt = 1 L | ν = frequency | u = speed

$m_p = 1.6726 \times 10^{-27} kg$ | $m_n = 1.675 \times 10^{-27} kg$ | $m_e = 9.109 \times 10^{-31} kg$

Periodic Table Group Numbers are 1 through 18 on the periodic table.
In an alternative system numbering system, each group has a number from 1 through 8 and either the letter A or B. The correspondence between the two numbering systems follows:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

1A 2A 3B 4B 5B 6B 7B 8B 8B 8B 1B 2B 3A 4A 5A 6A 7A 8A

SOLUBLE IONIC COMPOUNDS

Compounds containing: Group 1(1A) ions, NH_4^+, NO_3^-, CH_3COO^-, ClO_4^-

Cl^-, Br^-, I^- except compounds of Ag^+, Pb^{2+}, Cu^+, Hg_2^{2+}

F^- except compounds of Group 2(2A) ions and of Pb^{2+}

SO_4^{2-} except compounds of Ca^{2+}, Sr^{2+}, Ba^{2+}, Ag^+, Pb^{2+}

INSOLUBLE IONIC COMPOUNDS Compounds containing:

OH^- except compounds of Group 1(1A) ions and of Ca^{2+}, Sr^{2+}, Ba^{2+}

CO_3^{2-}, PO_4^{3-} except compounds of Group 1(1A) ions and of NH_4^+

S^{2-} except compounds of Group 1(1A) ions, of Group 2(2A) ions, and of NH_4^+

CHEMISTRY HALLIT EXAM II

Questions	V1	V2	V3	V4
1	E	A	D	B
2	D	A	A	A
3	B	A	D	B
4	D	E	D	B
5	C	D	E	D
6	A	D	E	C
7	B	A	C	C
8	D	B	A	D
9	A	B	D	B
10	C	D	C	D
11	D	C	D	E
12	E	C	D	D
13	D	D	B	B
14	A	C	A	B
15	E	B	C	A
16	C	E	B	C
17	A	A	A	D
18	B	A	D	A
19	B	D	A	E
20	C	D	C	A
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22	D	B	E	A
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159
Exam II
Fall 17

→ this key
V3